

100 Important CS Interview Questions with Answers

This PDF contains important interview questions with short and simple answers from Operating Systems, Computer Networks, Software Engineering, and Cloud Computing. Useful for placements, viva, and technical interviews.

Operating Systems

Q1. What is an Operating System?

Answer: An Operating System is system software that manages hardware and software resources.

Q2. What are the functions of an OS?

Answer: It manages memory, files, processes, devices, and security.

Q3. Difference between process and thread.

Answer: A process is an independent program while a thread is a lightweight unit inside a process.

Q4. What is multitasking?

Answer: Multitasking allows multiple tasks to run simultaneously.

Q5. What is context switching?

Answer: Context switching is saving one process state and loading another.

Q6. What is deadlock?

Answer: Deadlock occurs when processes wait forever for resources held by each other.

Q7. Necessary conditions for deadlock.

Answer: Mutual exclusion, hold and wait, no preemption, and circular wait.

Q8. What is starvation in OS?

Answer: Starvation happens when a process never gets CPU resources.

Q9. Difference between paging and segmentation.

Answer: Paging divides memory into fixed blocks while segmentation divides based on logical units.

Q10. What is virtual memory?

Answer: Virtual memory uses disk space as temporary RAM.

Q11. What is thrashing?

Answer: Thrashing occurs when excessive paging reduces performance.

Q12. What is demand paging?

Answer: Pages are loaded into memory only when needed.

Q13. What is a semaphore?

Answer: A semaphore is used for process synchronization.

Q14. What is mutual exclusion?

Answer: It ensures only one process accesses a resource at a time.

Q15. Difference between user mode and kernel mode.

Answer: User mode has limited access while kernel mode has full access.

Q16. What is CPU scheduling?

Answer: CPU scheduling decides which process gets CPU next.

Q17. Difference between preemptive and non-preemptive scheduling.

Answer: Preemptive scheduling interrupts processes while non-preemptive does not.

Q18. Explain FCFS scheduling.

Answer: Processes execute in the order they arrive.

Q19. Explain Round Robin scheduling.

Answer: Each process gets a fixed time slice in cyclic order.

Q20. What is fragmentation?

Answer: Unused memory spaces are called fragmentation.

Q21. Difference between logical and physical address.

Answer: Logical address is generated by CPU while physical address exists in RAM.

Q22. What is booting?

Answer: Booting is starting the computer and loading the OS.

Q23. Difference between multiprogramming and multitasking.

Answer: Multiprogramming keeps CPU busy while multitasking improves user interaction.

Q24. What are system calls?

Answer: System calls allow programs to request OS services.

Q25. What is a race condition?

Answer: Race condition occurs when multiple processes access shared data simultaneously.

Computer Networks

Q26. What is a computer network?

Answer: A computer network connects devices for communication and resource sharing.

Q27. What is the OSI model?

Answer: OSI is a 7-layer networking model for communication.

Q28. Explain TCP/IP model.

Answer: TCP/IP is a networking model with four layers used on the internet.

Q29. Difference between TCP and UDP.

Answer: TCP is reliable and connection-oriented while UDP is faster and connectionless.

Q30. What is an IP address?

Answer: An IP address uniquely identifies a device on a network.

Q31. Difference between IPv4 and IPv6.

Answer: IPv4 uses 32-bit addresses while IPv6 uses 128-bit addresses.

Q32. What is DNS?

Answer: DNS converts domain names into IP addresses.

Q33. What is DHCP?

Answer: DHCP automatically assigns IP addresses to devices.

Q34. What is a MAC address?

Answer: A MAC address is a unique hardware identifier for a network device.

Q35. What is subnetting?

Answer: Subnetting divides a network into smaller networks.

Q36. What is a router?

Answer: A router forwards data between networks.

Q37. What is a switch?

Answer: A switch connects devices within the same network.

Q38. Difference between hub and switch.

Answer: Hub broadcasts data while switch sends data to specific devices.

Q39. What is HTTP and HTTPS?

Answer: HTTPS is the secure version of HTTP using encryption.

Q40. What is a firewall?

Answer: A firewall protects networks from unauthorized access.

Q41. What is packet switching?

Answer: Data is divided into packets before transmission.

Q42. What is latency?

Answer: Latency is the delay in data transmission.

Q43. What is bandwidth?

Answer: Bandwidth is the amount of data transferred per second.

Q44. Explain three-way handshake.

Answer: TCP establishes connection using SYN, SYN-ACK, and ACK.

Q45. What is congestion control?

Answer: It prevents excessive traffic in a network.

Q46. Difference between public and private IP.

Answer: Public IP is internet accessible while private IP is used internally.

Q47. What is ARP?

Answer: ARP maps IP addresses to MAC addresses.

Q48. What is SMTP?

Answer: SMTP is used for sending emails.

Q49. What is FTP?

Answer: FTP transfers files between computers.

Q50. What is VPN?

Answer: VPN provides secure internet communication.

Software Engineering

Q51. What is Software Engineering?

Answer: Software Engineering is the systematic development of software.

Q52. What is SDLC?

Answer: SDLC is the Software Development Life Cycle.

Q53. Explain waterfall model.

Answer: Waterfall follows sequential development phases.

Q54. Explain Agile methodology.

Answer: Agile focuses on iterative and flexible development.

Q55. What is Scrum?

Answer: Scrum is an Agile framework using sprints.

Q56. Difference between verification and validation.

Answer: Verification checks process while validation checks product.

Q57. What is software testing?

Answer: Testing identifies software defects.

Q58. Difference between black box and white box testing.

Answer: Black box tests functionality while white box tests internal code.

Q59. What is unit testing?

Answer: Unit testing checks individual modules.

Q60. What is integration testing?

Answer: Integration testing checks combined modules.

Q61. What is regression testing?

Answer: Regression testing ensures old features still work.

Q62. What is debugging?

Answer: Debugging is fixing software errors.

Q63. What is requirement analysis?

Answer: Requirement analysis identifies customer needs.

Q64. What is feasibility study?

Answer: It checks whether a project is practical and achievable.

Q65. What is UML?

Answer: UML is a modeling language for software design.

Q66. What is a use case diagram?

Answer: It shows interactions between users and the system.

Q67. What is cohesion?

Answer: Cohesion measures how related module functions are.

Q68. What is coupling?

Answer: Coupling measures dependency between modules.

Q69. Difference between functional and non-functional requirements.

Answer: Functional requirements define features while non-functional define quality.

Q70. What is risk management?

Answer: Risk management identifies and handles project risks.

Q71. What is software maintenance?

Answer: Maintenance updates and fixes software after deployment.

Q72. What is version control?

Answer: Version control tracks code changes.

Q73. What is Git?

Answer: Git is a distributed version control system.

Q74. What is CI/CD?

Answer: CI/CD automates software integration and deployment.

Q75. Difference between prototype and spiral model.

Answer: Prototype builds sample models while spiral focuses on risk management.

Cloud Computing

Q76. What is cloud computing?

Answer: Cloud computing delivers computing services over the internet.

Q77. What are the advantages of cloud computing?

Answer: It provides scalability, flexibility, and cost savings.

Q78. What is virtualization?

Answer: Virtualization creates virtual versions of hardware resources.

Q79. Difference between IaaS, PaaS, and SaaS.

Answer: IaaS provides infrastructure, PaaS provides platform, SaaS provides software.

Q80. What is AWS?

Answer: AWS is Amazon's cloud computing platform.

Q81. What is Microsoft Azure?

Answer: Azure is Microsoft's cloud platform.

Q82. What is Google Cloud Platform?

Answer: GCP is Google's cloud service platform.

Q83. What is serverless computing?

Answer: Serverless runs applications without managing servers.

Q84. What is load balancing?

Answer: Load balancing distributes traffic across servers.

Q85. What is auto scaling?

Answer: Auto scaling automatically adjusts resources based on demand.

Q86. What is Docker?

Answer: Docker is a containerization platform.

Q87. What is Kubernetes?

Answer: Kubernetes manages containerized applications.

Q88. What is a virtual machine?

Answer: A VM is a software-based computer system.

Q89. Difference between containers and virtual machines.

Answer: Containers share OS while VMs have separate OS.

Q90. What is cloud storage?

Answer: Cloud storage stores data on remote servers.

Q91. What is elasticity in cloud computing?

Answer: Elasticity automatically adjusts resources dynamically.

Q92. What is scalability?

Answer: Scalability increases system capacity when needed.

Q93. What is edge computing?

Answer: Edge computing processes data near the source.

Q94. What is multi-cloud?

Answer: Multi-cloud uses services from multiple providers.

Q95. What is hybrid cloud?

Answer: Hybrid cloud combines public and private clouds.

Q96. Difference between public and private cloud.

Answer: Public cloud is shared while private cloud is dedicated.

Q97. What is cloud security?

Answer: Cloud security protects cloud systems and data.

Q98. What is CDN?

Answer: CDN delivers content faster using distributed servers.

Q99. What is DevOps?

Answer: DevOps combines development and operations practices.

Q100. What is Infrastructure as Code?

Answer: IaC manages infrastructure using code.